

# Disease Risk Mapping

Spatial data science for visualising disease risk

**Objective** — Demonstrate reproducible workflows for spatial disease risk mapping and geographic visualisation.

**Methods** — Spatial data handling, choropleth and point mapping, and exploratory spatial visualisation.

**Tools** — R (tidyverse, ggplot2, sf).

*This page is a methods demonstration and is being expanded.*

## Introduction

### Data

I start it with the penguins dataset from @palmerpenguins package [c@corsi2021]

```
library(tidyverse)
library(palmerpenguins)
library(gt)
glimpse(penguins)
```

Rows: 344

Columns: 8

```
$ species      <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island       <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g   <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex          <fct> male, female, female, NA, female, male, female, male~
$ year         <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

## Species

Figure 1 is a scatter plot of species of penguins

```
ggplot(penguins,  
  aes(x = bill_length_mm, y = bill_depth_mm,  
    color = species, shape = species)  
)+  
  geom_point()+  
  theme_minimal()+  
  #scale_color_continuous()+  
  labs(x= "Bill length (mm)", y="Bill depth (mm)")
```

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_point()`).

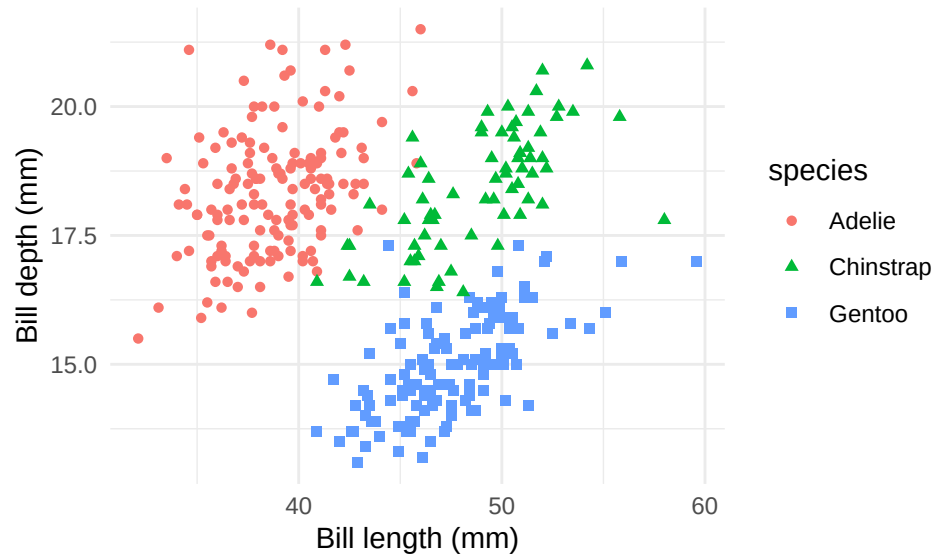


Figure 1: A scatter plot

## Penguins

Table 1 below shows first 10 penguins from the dataset.

Table 1: First 10 penguins

| species | island    | bill_length_mm | bill_depth_mm |
|---------|-----------|----------------|---------------|
| Adelie  | Torgersen | 39.1           | 18.7          |
| Adelie  | Torgersen | 39.5           | 17.4          |
| Adelie  | Torgersen | 40.3           | 18.0          |
| Adelie  | Torgersen | NA             | NA            |
| Adelie  | Torgersen | 36.7           | 19.3          |
| Adelie  | Torgersen | 39.3           | 20.6          |
| Adelie  | Torgersen | 38.9           | 17.8          |
| Adelie  | Torgersen | 39.2           | 19.6          |
| Adelie  | Torgersen | 34.1           | 18.1          |
| Adelie  | Torgersen | 42.0           | 20.2          |

```
penguins|>
  slice_head(n=10)|>
  select(species, island, bill_length_mm, bill_depth_mm)|>
  gt()
```

## Analysis

### modelling

### References